

Software engineering lab 03

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SECTION : K

LAB 03 : Component Modelling and architectural pattern selection

Scenario Review :

Project Title :

Personal Subscription & Recurring Expense Auditor

System Overview

The Personal Subscription & Recurring Expense Auditor analyzes transaction data to identify recurring billing patterns and calculate the user's monthly subscription expenditure. The system allows users to import transaction data in CSV or OFX format and provides information about recurring subscriptions, renewal dates, and monthly recurring expenses.

Key Functional Requirements

1. The system shall identify recurring billing patterns and estimate the total monthly subscription expenditure.
2. The system shall allow users to import transaction data in CSV or OFX format.
3. The system shall alert the user before the renewal date of an identified subscription.
4. The system shall calculate and display the user's total monthly recurring expense or burn rate.
5. The system shall provide a one-click cancellation guide for each identified subscription.

Key Non-Functional Requirements

1. Financial transaction data must be processed securely without storing raw banking credentials in cleartext.
2. The system shall provide a clear and easy-to-understand interface.
3. Users should be able to easily view subscription name, recurring amount, renewal date, and monthly burn rate.

Main Challenges

1. Security:

The system handles sensitive financial transaction information, so transaction data must be processed securely.

2. Performance:

The system needs to efficiently process transaction records and identify recurring billing patterns.

3. Usability:

The system should provide a simple and clear interface so that users can easily understand and manage their recurring expenses.

4. Data Processing:

The system must validate transaction data and identify recurring patterns based on factors such as merchant, amount, and billing frequency.

Conclusion

The scenario requires an architecture that can securely process financial transaction data, efficiently analyze recurring expenses, and provide a clear interface for users.

Step 2: Architectural Style Analysis

Three architectural styles were analyzed for the Personal Subscription & Recurring Expense Auditor system: Layered Architecture, Microservices Architecture, and Client-Server Architecture.

1. Layered Architecture

Layered Architecture organizes the system into separate layers such as Presentation Layer, Business Logic Layer, and Data Layer.

Advantages:

- Clear separation of concerns.
- Easy to understand and maintain.
- Suitable for traditional applications.

Disadvantages:

- Performance overhead due to communication between layers.
- Tight coupling between layers.
- Difficult to scale individual components independently.

2. Microservices Architecture

Microservices Architecture divides the system into small and independently deployable services.

Advantages:

- Independent scaling of services.
- Fault isolation.
- Independent deployment.

Disadvantages:

- Increased operational complexity.
- Network latency.
- Data consistency challenges.

3. Client-Server Architecture

Client-Server Architecture consists of multiple clients communicating with a centralized server.

Advantages:

- Centralized control.
- Simple deployment.
- Easy data consistency.

Disadvantages:

- Single point of failure.
- Scalability limitations.

- Limited fault tolerance.

Comparison

Architecture	Main Advantage	Main Disadvantage
Layered	Clear separation of concerns	Performance overhead
Microservices	Independent scaling	Operational complexity
Client-Server	Centralized control	Single point of failure

Conclusion

All three architectural styles can be used for the Personal Subscription & Recurring Expense Auditor system. The final architecture will be selected in Step 3 based on system requirements, security, performance, and maintainability.

Architecture Selection & Component Identification

Architecture Selection

We chose **Layered Architecture** for the Personal Subscription & Recurring Expense Auditor system.

Reasons for Selection

1. **Clear Separation of Responsibilities:**

The system can separate the user interface, transaction analysis and business logic, and transaction data management into different layers. This makes the system easier to understand and maintain.

2. **Suitable for the System Requirements:**

The system performs structured operations such as importing transaction data, identifying recurring subscriptions, calculating monthly expenses, and displaying the results to the user. These operations can be organized clearly across different layers.

Components Identified

1. **User Interface Component**

Provides the interface through which the user imports transaction files and views subscription and expense information.

2. **Transaction Import Component**

Handles importing and validating transaction data from CSV or OFX files.

3. Subscription Analysis Component

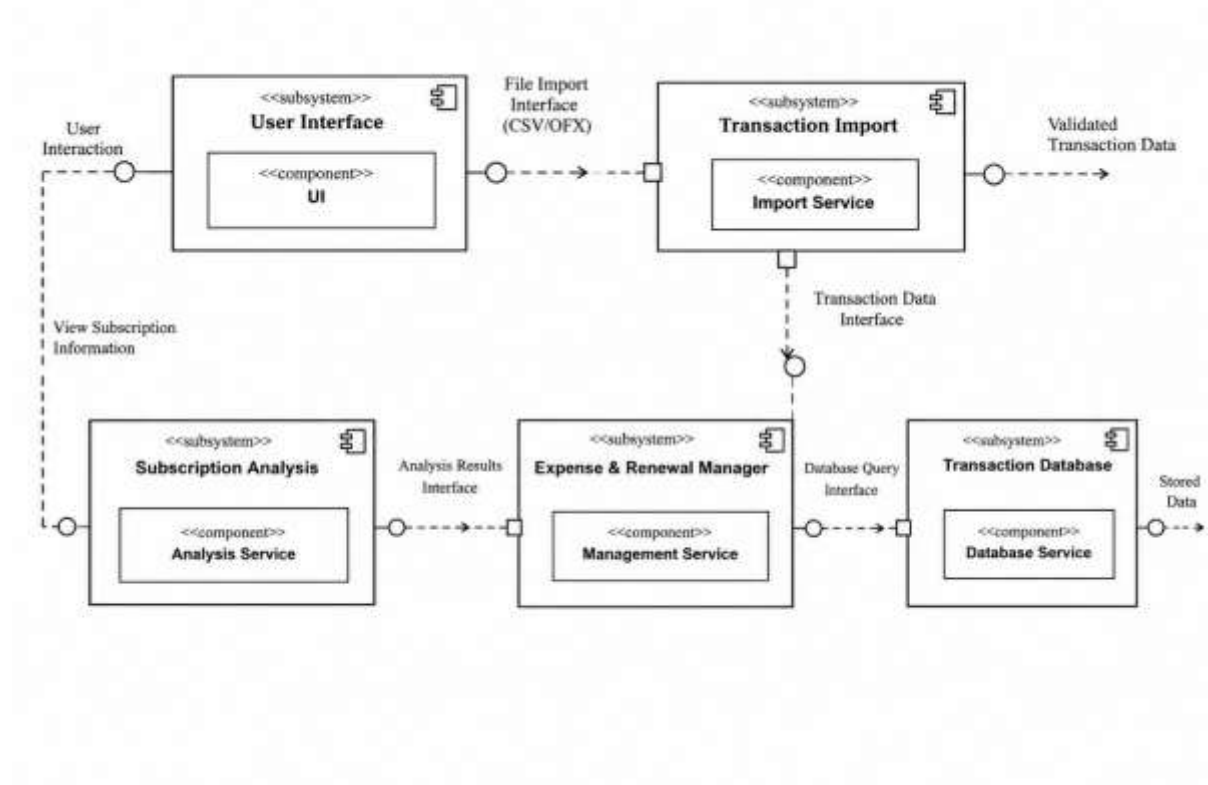
Analyzes transaction records and identifies recurring subscription patterns.

4. Expense & Renewal Manager Component

Calculates monthly recurring expenses, determines renewal dates, and manages renewal alerts.

5. Transaction Database Component

Stores the required transaction and subscription information used by the system.



Architecture Selection:

We chose **Layered Architecture** for the Personal Subscription & Recurring Expense Auditor system.

Reasons for Selection:

1. **Clear separation of responsibilities:** The system can separate the user interface, transaction processing and subscription analysis, and data management into different layers. This makes the system easier to understand, maintain, and modify.
2. **Suitable for the system requirements:** The architecture supports the main operations of the system, such as importing CSV/OFX transaction data, validating transactions, identifying recurring subscription patterns, calculating monthly recurring expenses, determining renewal dates, and displaying the results to the user.

Security Advantage:

Layered Architecture separates transaction processing and data management from the user interface. This helps control access to sensitive financial transaction data and supports secure processing of CSV/OFX imports without storing raw banking credentials in cleartext, as required by NFR-001.

Performance Benefit:

The system can process transaction data through separate layers, allowing validation, recurring-pattern analysis, and expense calculation to be handled in an organized manner. This supports efficient processing of transaction records and calculation of the monthly burn rate.

Conclusion:

Layered Architecture provides a clear and structured organization for the Personal Subscription & Recurring Expense Auditor and supports its functional, security, usability, and processing requirements.